DepthWizard — Demo Video Script

SIH 2026 · PS **SIH26175** (ISRO / SAC) · Theme: **Disaster Management**. A 5–6 minute screen recording of the local prototype, narrated live. Green boxes are what you **do**; bordered boxes are what you **say**.

Before you hit record

Step	Why
Start the app: `cd heimdall-web && npm run dev`	Wait until it says Ready. Open `localhost:3000`.
Pre-run one job and leave it in Recent jobs	Insurance. If the live run is slow you can cut to a finished one without dead air.
Have `Heimdall_test_samples/AHN/utrecht_suburb__UPLOAD_THIS.tif` ready	This is the file you drag in on camera.
Browser at 100% zoom, full screen, close other tabs	The 3D view needs the space and a clean frame reads as professional.
Leave the GSD override empty	It only applies to images with no coordinate system. A leftover value overrides a GeoTIFF's own scale and raises a warning banner mid-take.
Click into a scene once and orbit, then reload	Warms the WebGL context so the first render is not janky on camera.
Silence notifications	An ntfy popup mid-take costs you a re-record.

> **One rule for the whole video: never read a number off the screen that you have not earned.** Every figure below is measured and in the repository. If an evaluator pauses the video and checks, it holds up — that is the impression you want.

The shape of it

Time	Beat	What is on screen
0:00–0:35	The problem	Landing page, one bundled scene already open
0:35–1:20	Upload a real image, start it	Drag the GeoTIFF in, press Process image
1:20–2:20	How it works, while it runs	Live progress log and stage list
2:20–3:20	The result — fly it	3D flythrough, Orbit → Fly, Measure
3:20–4:20	**The proof** — validate the job you just ran	Upload the LiDAR, read the metrics, toggle Heimdall ↔ Reference
4:20–5:20	The numbers	Accuracy tab
5:20–6:00	Honest limits, then close	Back to the 3D view

0:00 – 0:35 · The problem

Do: Open on the app with **Utrecht — suburban** already loaded in the 3D view. Slowly orbit once while you talk. Do not click anything yet.

SAY

Elevation data is the first thing you want after a flood or an earthquake, and the first thing you do not have. A LiDAR survey is a commissioned flight — weeks and a budget. What you usually *do* have is a satellite photograph.

This is DepthWizard. One ordinary RGB image in; the surface in metres out — buildings, terrain, canopy — as a 3D scene you can fly through. No stereo pair, no LiDAR.

Do not say "AI-powered" or "state of the art". The people watching build satellites. Show them a measurement and they will trust you; use marketing words and they will not.

0:35 – 1:20 · Upload something real

Do: Click **New processing job**. Drag `utrecht_suburb__UPLOAD_THIS.tif` onto the drop zone. Let the thumbnail appear. Press **Process image**. Let the progress bar start moving.
Check first: the **GSD override** box under Advanced must be **empty**. A GeoTIFF states its own scale; a leftover value replaces it and the app will show a warning banner on camera.

SAY

A real aerial GeoTIFF of a suburb near Utrecht, 400 metres across at 25 centimetres per pixel — one of eight sites we hold out, never seen in training.

No preprocessing, no parameters. The file carries its own coordinate system, so the pipeline reads the scale from the image itself.

Do: While it runs, hover the **Reference DEM** and **Ground control points** fields so they are visible on camera, but leave them empty.

SAY

Both optional. Supply a local terrain model or surveyed control points and the output ties to your datum — one control point is worth about half a metre on absolute elevation. I am leaving them empty to show the default path: everything it needs, it fetches itself.

1:20 – 2:20 · How it works, while it runs

Do: Point at the live stage list as each one lights up — **Ingest & georeferencing**, **Ground sample distance**, **Height estimation**, **Absolute calibration**, **Textured 3D mesh**.

SAY

While that runs, the interesting part is why this works at all.

A depth model trained on ordinary photographs predicts *relative* depth — what is nearer, what is further. It has no

idea what a metre is. So we freeze that model and train a small head on top of it, against real airborne LiDAR, whose only job is to turn relative depth into metres above ground.

The obvious alternative is a metric depth model that outputs metres directly. We tried it. Those models assume a camera with a focal point, and a satellite orthomosaic has none — every pixel looks straight down. Its correlation with true height was **0.06**. Effectively zero. The relative model reaches **0.41** before we train anything on top.

SAY

[OPTIONAL — cut this first if you are running long]
That one measurement is why the system is built this way. And it matches what the problem statement itself suggests: anchoring scale-agnostic depth to a coarse DEM.

If the job finishes early, stop talking and move on. Do not stretch the explanation to fill time. A crisp video that ends at 5:20 beats a padded one at 6:30.

2:20 – 3:20 · The result

Do: The job completes and the **3D flythrough** opens. Orbit once. Switch to **Fly** and move slowly down a street at roof height. Keep it smooth — no fast spins.

SAY

That is the surface, from the single image I uploaded ten seconds ago. True metres, no vertical exaggeration.

Do: Click **Measure**. Draw a line across two or three buildings. Let the elevation profile appear underneath.

SAY

Distance along the ground, and the height profile underneath it.

Do: Enable the **flood** toggle and drag the water level up a few metres. Let the water plane cut through the scene.

SAY

And because the theme here is disaster management — this cuts a water plane through the scene at any level and shows what goes under. From one photograph, on a laptop, in seconds.

Do: Turn the flood off before moving on.

3:20 – 4:20 · The proof

> **This is the most important minute of the video.** Anyone can show a plausible-looking 3D render. Almost nobody shows it next to the real measurement. Slow down here.

> **Do:** Scroll to **Validate against reference** under the viewer. Upload `utrecht_suburb__truth_heights.tif`. Wait for the metrics table and error map. A **Compare against the reference** toggle appears above it — flip **Heimdall ↔ Reference** two or three times, slowly, from the same camera angle.

SAY

This is the part I would want to see if I were evaluating this.

I am handing it the airborne LiDAR for the same ground — the real measurement — and it scores itself against it. Two and a half metres on height above ground, and it draws where the error is.

And this is the same surface I just reconstructed, flipped against the truth. Same grid, same texture, same vertical origin, so only the heights differ. Roads, gardens, the ground between the houses land within about twenty centimetres.

> **Do:** Open **Example scenes** and click **Philadelphia — riverside**, then toggle **Heimdall ↔ LiDAR truth** once.

SAY

And a US city from GAMUS, the dataset the problem statement recommends — so this is not one country or one sensor.

4:20 – 5:20 · The numbers

> **Do:** Click the **Accuracy** tab. Let the table fill the frame. Point at the mean row, then at the per-landscape rows.

SAY

Every number behind that comparison is in the app, not in a slide.

Across eight held-out Dutch sites, scored against airborne LiDAR: **5.32 metres RMSE** on height above ground, correlation **0.67**. The honest reference point is predicting nothing at all — assuming flat ground everywhere — and that is **8.37 metres**.

On the full absolute DSM, terrain plus structures, we are at **6.11 metres** — and about half a metre of that is not our model at all. It is the constant offset between the global terrain datum and the local one. Supply a single ground control point and it drops to **5.63**.

Ground and low vegetation, which are most of any image, come in at **0.82** and **0.86** metres.

> **Do:** Point at the four landscape rows.

SAY

The problem statement asks for stability across urban, sparse, hilly and forested terrain, so we measured each separately rather than hiding behind one average. Forest 2.24, sparse farmland 1.56, hilly town 3.21, urban 8.40.

> **Do:** Scroll to the resolution figures.

SAY

And because evaluation uses Cartosat 2S at sixty centimetres — the same model, on the same sites, with the imagery degraded to match. **5.35** metres at twenty-five centimetres, **5.51** at sixty, **5.72** at one metre. Essentially flat across the range that matters.

Past two metres it degrades, and by ten metres a building is one to three pixels across and there is nothing left to measure. We say so, and the pipeline warns you when the input is outside the range it was trained for.

5:20 – 6:00 · Limits, then close

> **Do:** Go back to the **3D flythrough** on a scene with tall buildings. Orbit gently while you finish.

SAY

One limitation, stated plainly, because you will see it if you look closely.

Tall structures come out short — roughly half their true height on the tallest buildings. A single nadir photograph contains no parallax, so when the model cannot tell a ten-metre building from a twenty-metre one, predicting something in between is mathematically the right answer. We could scale it up to look better and the measured error would get worse, so we do not.

Ground and roads stay sub-metre. Buildings and trees carry almost all the remaining error, and we publish exactly how much.

SAY

All of this runs on a laptop from one command. The trained head is six and a half megabytes, and every figure I quoted regenerates from a script in the repository.

One photograph in — a measured, explorable 3D surface out. Thank you.

Delivery notes

- **Pace.** 885 spoken words — **5 min 54 at a normal pace, 6 min 48 if you speak slowly.** Read it once with a timer first. If you run long, the block marked OPTIONAL at 1:20 is the one to cut.
- **Mouse.** Move deliberately. Hover a control for a beat before clicking it so the viewer's eye can follow.
- **The two moments that win it** are the ground-truth toggle at 3:20 and the honest limitation at 5:20. Slow down for both. Volunteering a weakness reads as confidence, and evaluators notice.
- **If the live job fails on camera**, do not restart it. Say "I have one already processed" and click it in **Recent jobs**. Recovering smoothly looks better than a clean take.
- **Do not** show the terminal, the code, or the file system. Prototype only.

Numbers, for quick reference

Figure	Value
Mean RMSE, 8 held-out LiDAR sites	**5.32 m**
Mean absolute error · correlation	3.04 m · 0.67
Predicting zero height (the baseline we beat)	8.37 m
Ground · low vegetation	0.82 m · 0.86 m
Forest · sparse · hilly · urban	2.24 · 1.56 · 3.21 · 8.40 m
Resolution: 0.25 m → 1 m → 2 m	5.35 → 5.72 → 6.44 m
GAMUS test split (703 tiles)	4.74 m, vs 8.59 m predicting nothing
Metric backbone vs relative, correlation with true height	0.06 vs 0.41
Trained head size · inference	6.6 MB · ~4.5 s per 400 m site

Every figure regenerates with `python scripts/benchmark_ahn.py` and is read from `outputs/benchmark_ahn/results.json`.
Repository: github.com/Aryan241/Heimdall